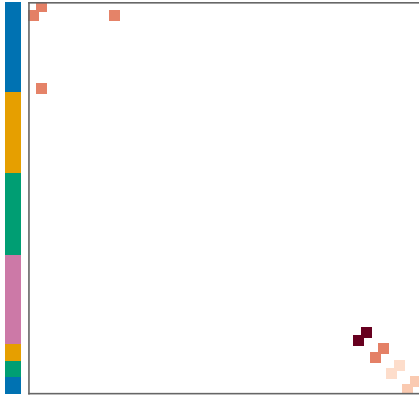


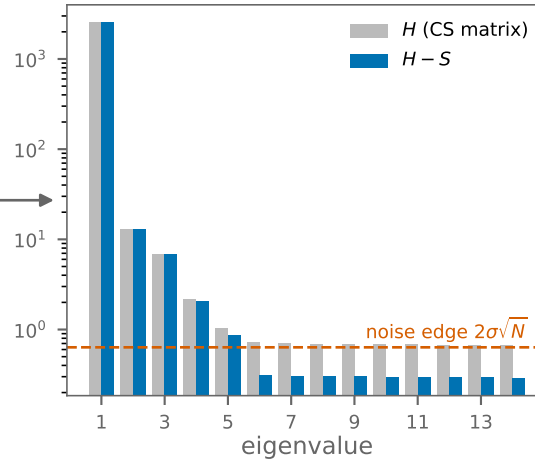
Large N ($N \leq 20,000$): dwg (detection + whitening, shown on the CS scale)

--robust (auto); operator engine

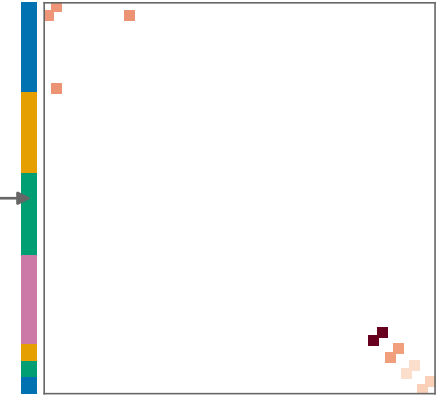
1 pass 1 + KING over all pairs
(popcounts): candidates > 0.04



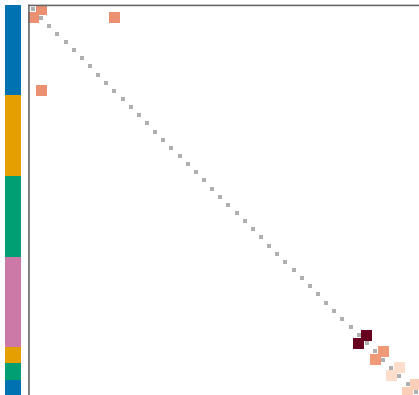
2 rank from the noise edge
 $N = 2000$: rank 5



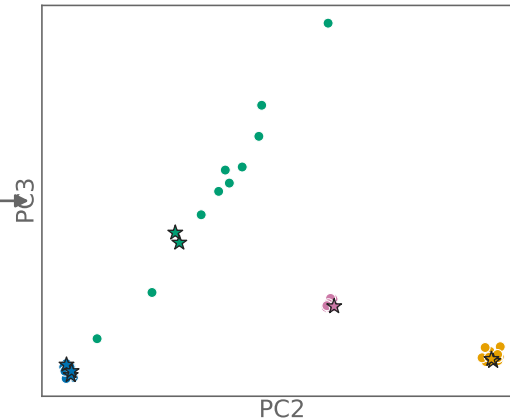
3 fit $H \approx L + S$: pairs with
 $\hat{\phi} > \tau$ among the candidates



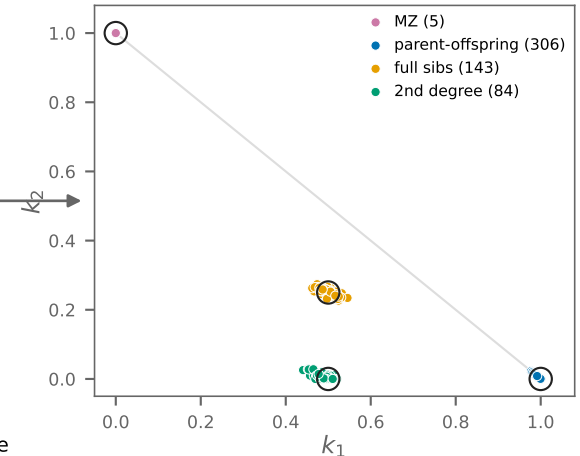
4 $\Sigma = v^{1/2}(I + 2\Psi)v^{1/2}$
block diagonal over families



5 PCs of $\Sigma^{-1/2}G$,
mapped back by $\Sigma^{1/2}$



6 final relatedness (1 pass)
 $N = 2000$, 538 pairs



● CEU ● CHB ● MXL ● YRI ★ relative

KING over all pairs: $N^2M/64$ popcounts (6.5 s at $N = 20,000$). Fit: each iteration one pass $G(G'V)$, nothing $N \times N$.
dwg at $N = 20,000$: 26.5 s in-core (standard PCAone 24 s). Panels 1, 3-5: the $N = 48$ example (readable); 2 and 6: simulated $N = 2000$.